

PAPER_163 UQFF Modular Architecture: Decomposed Compressed MUGE Functions

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Abstract

This paper establishes the **modular decomposition** of the 9-term Compressed MUGE into individual callable functions, each representing a single physical correction term. This transforms the monolithic `compute_compressed_MUGE()` function (SOURCE4) into an auditable, unit-testable architecture where each correction can be independently validated against observational data. Eight decomposed functions are cataloged with their governing equations and parameter dependencies.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4$ day¹, [SSq] = 0.57) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Original 9-Term Compressed MUGE (PAPER_090)

$$g_{comp}(r, t) = g_0 \cdot (1 + H_0 t) \cdot (1 - B/B_{crit}) \cdot f_{env}(r) \\ + \Lambda c^2/3 + \frac{\hbar}{\Delta x \Delta p} \cdot \int \psi \cdot \frac{2\pi}{t_H} \\ + \rho_{fluid} V g_{loc} + (M + M_{DM}) \cdot \left(\frac{\delta \rho}{\rho} + \frac{3GM}{r^3} \right)$$

2. Decomposed Function Architecture

Function 1: Base Newtonian Gravity

$$g_{base}(M, r) = \frac{G \cdot M}{r^2}$$

```
double compute_compressed_base(const MUGESystem& sys) {  
    return G * sys.M / (sys.r * sys.r);  
}
```

Function 2: Hubble Expansion Correction

$$g_{exp}(t) = 1 + H_0 \cdot t$$

$$H_0 = 67.4 \text{ km/s/Mpc} = 2.185 \times 10^{-18} \text{ s}^{-1}$$

```
double compute_compressed_expansion(const MUGESystem& sys) {
    const double H0 = 2.185e-18; // s^-1 (Planck 2018)
    return 1.0 + H0 * sys.t;
}
```

Function 3: Magnetic Suppression Adjustment

$$f_{super} = 1 - B/B_{crit}$$

```
double compute_compressed_super_adj(const MUGESystem& sys) {
    return 1.0 - sys.B / sys.B_crit;
}
```

Function 4: Envelope Confinement

$$f_{env}(r) = \begin{cases} 1 & r \leq R_{env} \\ e^{-(r-R_{env})/R_{env}} & r > R_{env} \end{cases}$$

```
double compute_compressed_envelope(const MUGESystem& sys) {
    if (sys.r <= sys.R_env)
        return 1.0;
    return std::exp(-(sys.r - sys.R_env) / sys.R_env);
}
```

Function 5: Cosmological Constant Term

$$g_{cosm} = \frac{\Lambda c^2}{3}$$

$$\Lambda = 1.1 \times 10^{-52} \text{ m}^{-2}$$

```
double compute_compressed_cosm(double Lambda = 1.1e-52) {
    const double c = 2.998e8;
    return Lambda * c * c / 3.0;
}
```

Function 6: Quantum Uncertainty Term

$$g_{quantum} = \frac{\hbar}{\Delta x \Delta p} \cdot \int \psi \cdot \frac{2\pi}{t_H}$$

where $t_H = 1/H_0$ is the Hubble time and $\int \psi$ is the integrated wave function amplitude.

```
double compute_compressed_quantum(double Delta_x, double Delta_p, double psi_int) {
    const double hbar = 1.055e-34;
    const double H0 = 2.185e-18;
    const double t_H = 1.0 / H0;
}
```

```

    return (hbar / (Delta_x * Delta_p)) * psi_int * (2.0 * M_PI / t_H);
}

```

Function 7: Buoyant Fluid Term

$$g_{fluid} = \rho_{fluid} \cdot V_{body} \cdot g_{local}$$

```

double compute_compressed_fluid(const MUGESystem& sys) {
    return sys.rho_fluid * sys.V_body * sys.g_local;
}

```

Function 8: Dark Matter Perturbation

$$g_{pert} = (M + M_{DM}) \cdot \left(\frac{\delta\rho}{\rho} + \frac{3GM}{r^3} \right)$$

```

double compute_compressed_perturbation(const MUGESystem& sys) {
    double tidal = 3.0 * G * sys.M / (sys.r * sys.r * sys.r);
    return (sys.M + sys.M_DM) * (sys.delta_rho_over_rho + tidal);
}

```

3. Full Decomposed Master Function

```

double compute_compressed_MUGE_modular(const MUGESystem& sys,
                                         double Delta_x = 1e-10,
                                         double Delta_p = 1e-24,
                                         double psi_int = 1.0) {
    double g_base    = compute_compressed_base(sys);
    double g_exp     = compute_compressed_expansion(sys);
    double g_super   = compute_compressed_super_adj(sys);
    double g_env     = compute_compressed_envelope(sys);
    double g_cosm    = compute_compressed_cosm();
    double g_quant   = compute_compressed_quantum(Delta_x, Delta_p, psi_int);
    double g_fluid   = compute_compressed_fluid(sys);
    double g_pert    = compute_compressed_perturbation(sys);

    return g_base * g_exp * g_super * g_env + g_cosm + g_quant + g_fluid + g_pert;
}

```

4. Unit Test Matrix

Function	Test Input	Expected Output	Tolerance
compute_base	M=1e30, r=1e11	6.67Ã10â	1%
compute_expansion	t=0	1.0000	<0.01%

Function	Test Input	Expected Output	Tolerance
compute_super_adj	$B=0$	1.0000	exact
compute_super_adj	$B=B_{\text{crit}}$	0.0000	exact
compute_cosm	$\hat{I}=1.1\text{e-}52$	$3.293\hat{A}10\hat{A}^3\hat{A}$	1%
compute_quantum	$\hat{I}\times\hat{I}_p=\hat{A}/2, \hat{I}=1$	varies	order of mag.
compute_fluid	$\hat{I}=1.29, V=1, g=9.81$	12.66	1%
compute_perturbation	$\hat{I}'\hat{I}/\hat{I}=0.1, M_{\text{DM}}/M=10$	varies	order of mag.

5. CP Integration

CP1 (CondensedPhysics.py): Add CompressedMUGEModularCalculator with 8 sub-methods. **CP2 (CondensedPhysics2.py):** Refactor existing compute_compressed_MUGE into modular form.

Status: $\hat{A}$ Complete | **CP Stage:** CP1/CP2 **Supersedes:** N/A (extends PAPER_090) | **Related:** PAPER_090 (9-term compressed), PAPER_158 (hybrid blending uses g_{comp}), PAPER_164 (quantum term calibration from CERN)